



Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

TANVI RAJESH PODUTWAR & 17/08/2026



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Session	2026-27 (ODD)	Course Name	LAB: Software Engineering
Semester	7 th	Course Code	23CT1704
Roll No	26	Name of Student	Tanvi Podutwar

Practical Number	Practical No.5
Course Outcome	Students will be able to understand the concepts and components of UML Class Diagrams. Students will be able to design and represent a Class Diagram for a given case study.
Aim	To study and draw UML Class diagram for given case Study.
Problem Definition	The objective is to study and design a UML Class Diagram for a given case study. The diagram represents the classes, attributes, methods, and relationships between different objects in the system. It helps in understanding the structure and functionality of the system before implementation.
Theory (100 words)	Theory A UML (Unified Modeling Language) Class Diagram is a structural diagram used to represent the static structure of a software system. It shows the different classes, attributes, methods, and relationships between classes. A class is represented using a rectangle divided into three sections: class name, attributes, and operations or methods. Class diagrams help developers understand how different objects in a system are connected and how they interact with each other. The main relationships used in a class diagram are association, aggregation, composition, inheritance, and dependency . Visibility symbols such as + for public, - for private, and # for protected can also be used. Class diagrams are useful during software analysis and design because they provide a clear blueprint of the system. They help in identifying system components, defining relationships, reducing design errors, and making the implementation process easier. Thus, UML Class Diagrams are an important tool for object-oriented software development.

UML Theory

UML (Unified Modeling Language) is a standard visual modeling language used to design, visualize, specify, and document software systems. It provides different diagrams to represent the structure and behavior of a system.

UML is mainly used in Object-Oriented Analysis and Design (OOAD). It helps developers and users understand the system before actual coding. UML diagrams are broadly divided into Structural Diagrams and Behavioral Diagrams.

- Structural diagrams show the static structure of a system, such as Class, Object, Component, and Deployment diagrams.
- Behavioral diagrams show the dynamic behavior of a system, such as Use Case, Sequence, Activity, and State diagrams.

A Class Diagram represents classes, attributes, methods, and relationships between classes. A Use Case Diagram represents interactions between users and the system.

Thus, UML provides a common and easy-to-understand way to model software systems and improves communication between developers, designers, and users.

1. **Structural Diagrams** – Represent the static structure of the system. Examples include **Class Diagram, Object Diagram, Component Diagram, and Deployment Diagram**.
2. **Behavioral Diagrams** – Represent the dynamic behavior of the system. Examples include **Use Case Diagram, Sequence Diagram, Activity Diagram, and State Diagram**.

A **Class Diagram** shows classes, attributes, methods, and relationships such as association, inheritance, aggregation, and composition. A **Use Case Diagram** shows the interaction between users (actors) and the system.

UML helps to **reduce development errors, improve communication, simplify complex systems, and provide a clear blueprint for software development**.

It is widely used during the analysis, design, and documentation phases of software development.



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Procedure and Execution (100 Words)	Algorithm / Steps in Detail 1. Start the process. 2. Study the given case study carefully and understand the system requirements, users, activities, and data involved. 3. Identify the main entities in the system. These entities will become the classes in the UML Class Diagram. 4. Identify the attributes of each class. Attributes represent the properties or data stored by a class, such as name, id, email, price, etc. 5. Identify the methods/operations of each class. These represent the actions performed by the class, such as <code>login()</code>, <code>register()</code>, <code>placeOrder()</code>, or <code>makePayment()</code>. 6. Determine the relationships between the classes. Identify whether the relationship is association, inheritance, aggregation, composition, or dependency. 7. Define multiplicity for relationships, such as 1, 0..1, 1..*, or *, to show how many objects are associated with another object. 8. Define visibility of attributes and methods using UML symbols: ○ + Public ○ - Private ○ # Protected 9. Draw each class using a rectangle divided into three sections: Class Name, Attributes, and Methods. 10. Connect the classes using the appropriate UML relationship lines and symbols. 11. Review the diagram and check whether all important classes, attributes, methods, relationships, and multiplicities are correctly represented. 12. Finalize and save the UML Class Diagram in the required format. 13. Stop.
	Output:



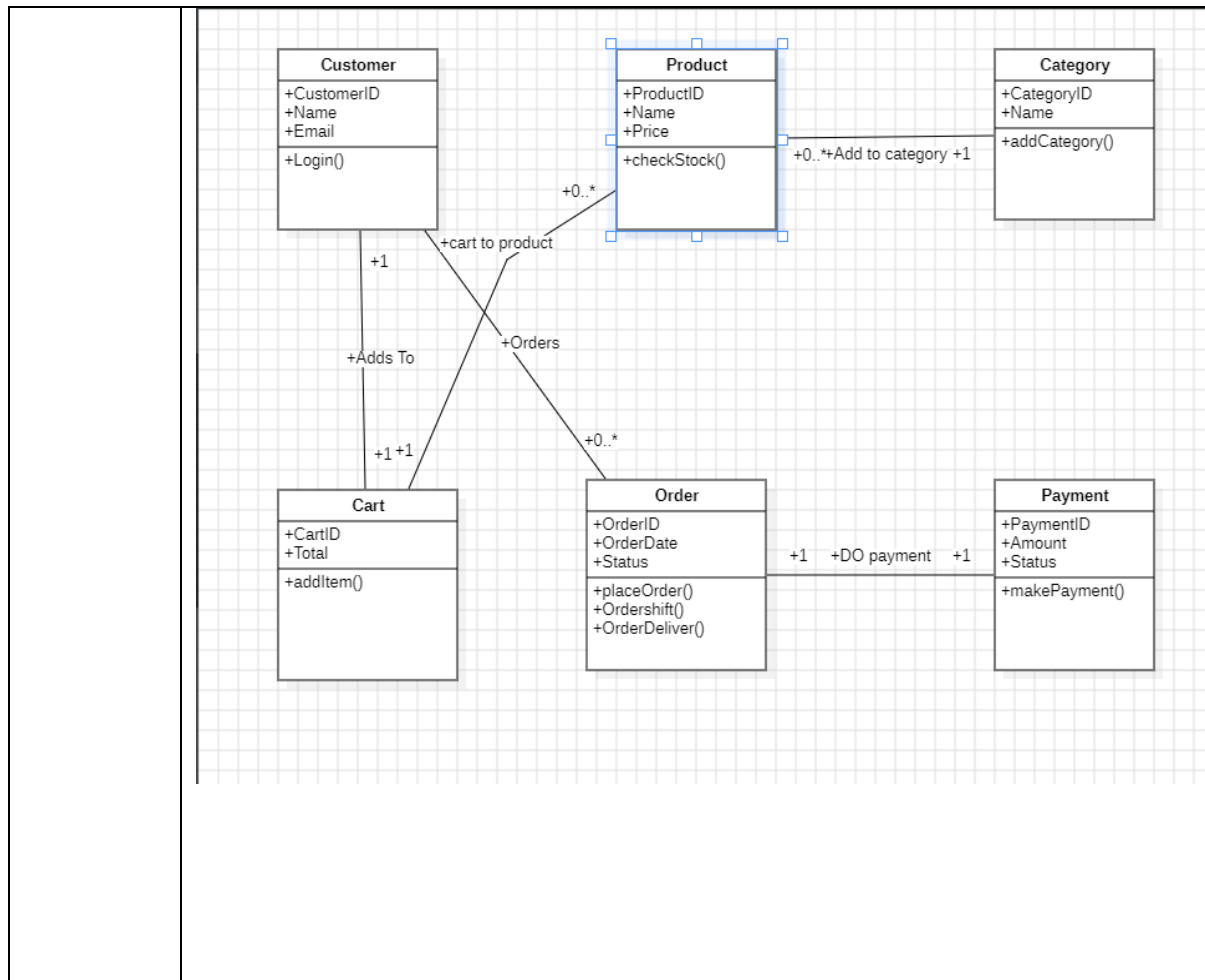
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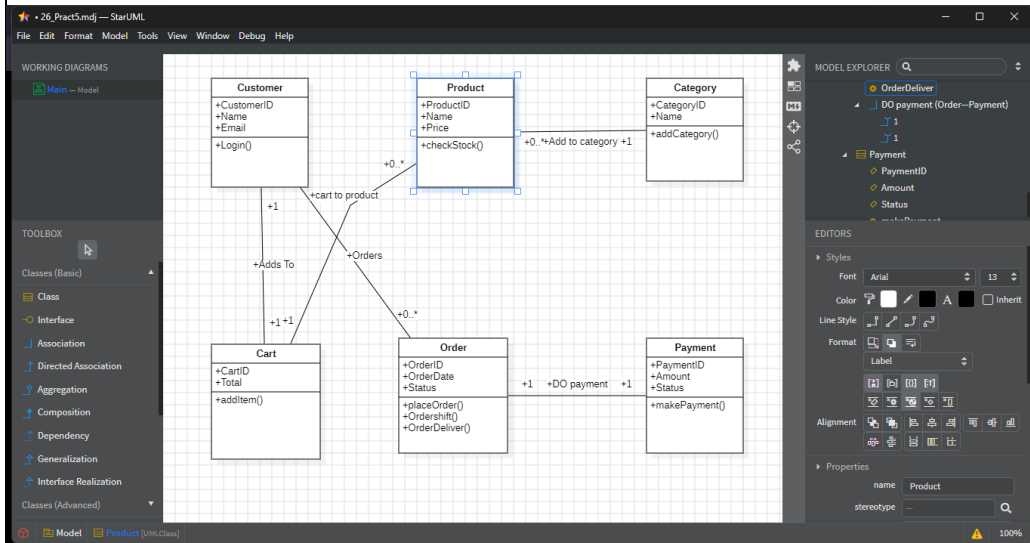
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Output Analysis

The output shows the UML Class Diagram for the E-Commerce Business Management System created using StarUML. The diagram contains the main classes: Customer, Product, Category, Cart, Order, and Payment.

- Customer can add products to the cart and place orders.
- Product contains product details such as Product ID, Name, and Price.
- Category is used to classify products.
- Cart stores selected products and maintains the total amount.
- Order manages order details and delivery information.
- Payment handles payment details and payment processing.

The relationships and multiplicities between the classes clearly represent how different components of the e-commerce system interact. Thus, the designed Class Diagram successfully represents the structure and relationships of the E-Commerce Business Management System.

1.



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Link of student Github profile where lab assignment has been uploaded	https://github.com/podutwartanvill11/software-engineering-lab/tree/main/Practical-5?utm_source=chatgpt.com
Conclusion	The practical was successfully completed by studying the given case study and applying the required Software Engineering concepts and techniques . The required diagram/model was designed correctly, providing a clear understanding of the system structure, components, and relationships. Thus, the objectives of Practical 5 were successfully achieved.
Plag Report (Similarity index < 12%)	



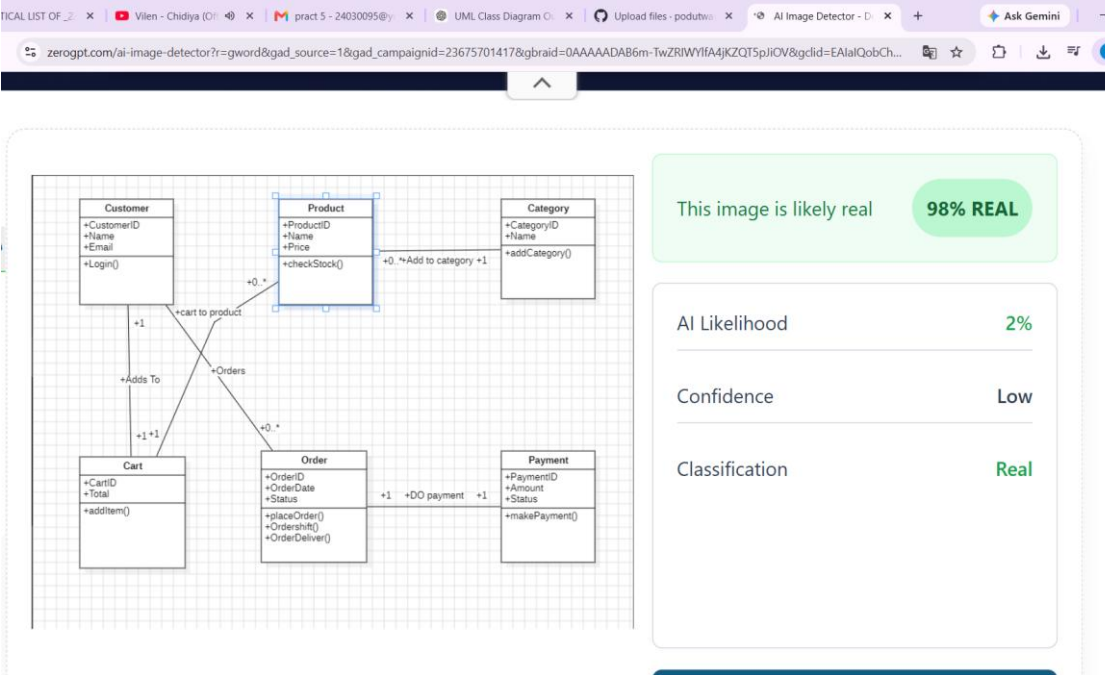
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Date	17/8/25